Centralized Document Management Information System (CDMIS)

Product Requirements Document (PRD) — University Archives Edition

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Target Audience:	University Records & Archives Offices, Academic Administrators, Technical Evaluators
Compliance Basis:	National Archives of the Philippines (NAP Form 1 - Records Inventory & Appraisal)
Deployment Mode:	Client-Side Persistent Single Page Application (Zero-Backend LocalStorage Engine)

1. Executive Summary

The Centralized Document Management Information System (CDMIS) is an institutional-grade records management and archiving portal. Originally architected for university records and archives offices, the system automates records classification, appraisal, retention scheduling, and access control in compliance with official National Archives standards (specifically modeled on NAP Form 1).

This application delivers an anonymized, high-fidelity portfolio demonstration showcasing complex enterprise workflows, multi-tiered Role-Based Access Control (RBAC), fine-grained document request/approval cycles, and zero-backend persistent state storage.

CDMIS PLATFORM ARCHITECTURE AT A GLANCE

• Foundation: Role-Based Shell (Admin / DRC / Staff) !' NAP Form 1 Ingestion Engine !' Access Request Pipeline
• Capabilities: Interactive Document Viewer (PDF/Image Zoom) | NAP Form 1 Excel Generator | Immutable Audit Logs

2. Problem Statement & Business Drivers

2.1 The Traditional Challenge

- **Fragmented Record Silos:** University documents remain scattered across unindexed physical file drawers and departmental shared drives.
- **Compliance Deficits:** Manual preparation of National Archives appraisal forms (NAP Form 1) leads to retention miscalculations and missing records.
- **Security Vulnerabilities:** Absence of digital restrictions leaves confidential personnel, fiscal, and legal files exposed without authorization checkpoints.
- **Opaque Auditing:** Paper-based records transfers lack immutable event tracking, obscuring who accessed, reviewed, or modified critical records.

2.2 CDMIS Solution Value

- **Standardized Ingestion:** Guided 4-section modal captures complete archival provenance, time/utility values, and disposition directives.
- **Automated Retention Computation:** System automatically validates and calculates Active + Storage = Total Retention Years.
- **Hierarchical Access Workflows:** Restricts confidential files behind an identity-verified access request and custodian approval pipeline.
- **Turnkey Portability:** Runs as a zero-backend client-persistent web app, allowing instant technical review without database setup.

3. User Personas & Role Matrix

The system enforces a strict 3-tier Role-Based Access Control (RBAC) hierarchy. Each user role operates within an isolated interface shell with tailored permissions and dashboard views.

VISUAL DIAGRAM: USER PERSONAS & INSTITUTIONAL CAPABILITIES



3.1 Persona Profiles & Access Scopes

Role	Persona Name	Primary Objectives	Key Pain Points Addressed
Administrator	Records Director	Global user/dept governance, cross-dept audits, compliance review	Manual compliance oversight; lack of centralized reporting
Custodian (DRC)	College Secretary	Department document ingestion, NAP appraisal, request approvals	Retention calculation complexity; lost paper requests
Staff Member	Faculty / Staff	Browse directory, discover policies, request confidential archives.	Long delays in obtaining archival records permissions.

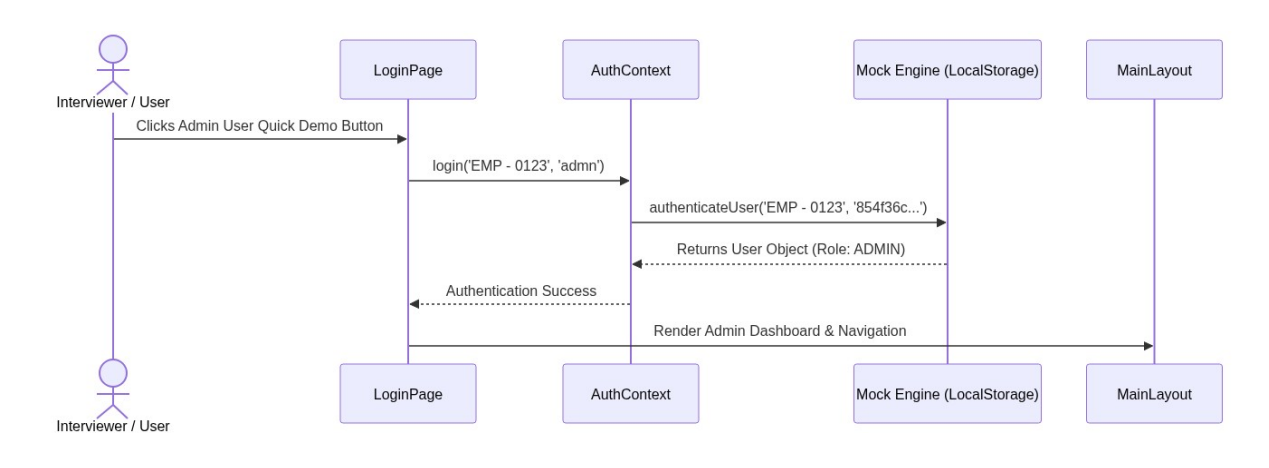
4. Feature Specifications & Requirements

4.1 Authentication & RBAC Engine

The authentication subsystem supports both simulated enterprise employee credentials and instant one-click evaluators:

- **Employee ID Scheme:** Formatted user identifiers matching standard institution patterns (^EMP - \d{4}$).
- **Simulated Hash Verification:** Emulates MD5 cryptographic digests for zero-backend verification.
- **One-Click Demo Suite:** Dedicated quick-login buttons for Admin, IT Custodian, HR Custodian, and Staff members.

VISUAL DIAGRAM: AUTHENTICATION & QUICK DEMO SESSION SEQUENCE FLOW



4.2 NAP Form 1 Records Inventory & Appraisal Engine

Implements the official National Archives of the Philippines Form 1 standards across four structured ingestion phases:

- **Section 1 - Office Information:** Office name, department, telephone, section, email, address, person-in-charge, and preparation date.
- **Section 2 - Record Series Details:** Title, description, inclusive dates covered, volume, medium, restrictions, filing location, frequency of use, and duplication.
- **Section 3 - Appraisal & Valuation:** Time Value (Temporary/Permanent), Utility Values (Administrative, Fiscal, Legal, Archival), Active/Storage retention periods, and disposition provision.
- **Section 4 - Digital Attachment & Classification:** File upload (PDF/TIFF up to 10MB), document type category, restriction classification (Public/Confidential), and comma-separated meta tags.

4.3 Interactive In-App Document Viewer & Inspection Modal

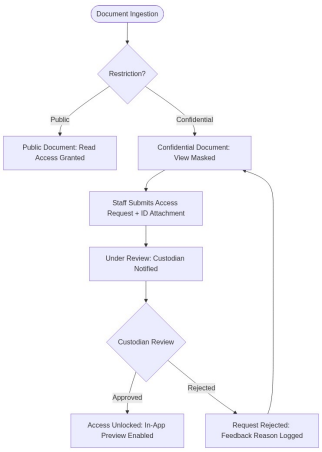
Enables browser-native inspection of archival assets without external software dependencies:

- **Multi-Format Canvas:** Native PDF rendering via iframe alongside high-resolution image scaling.
- **Zoom Controls:** Granular scaling from 50% to 300% with 1:1 reset and new-tab breakout.
- **Metadata Inspector Drawer:** Persistent side panel presenting uploader identity, timestamps, type categories, and technical attributes.

4.4 Access Control & Confidential Request Pipeline

Confidential documents are masked from unauthorized viewing. Users file digital access requests routed to the corresponding department custodian:

VISUAL DIAGRAM: CONFIDENTIAL DOCUMENT REQUEST LIFECYCLE WORKFLOW



4.5 Document Versioning & Audit History

Tracks document revisions with automated version incrementing (v1, v2, v3), complete commit notes, and historical rollback viewports.

4.6 Reporting & NAP Form 1 Excel Export

Features a client-side XML/Blob generator that converts repository metadata into official National Archives spreadsheet templates complete with three-tier signatory blocks.

4.7 Audit Logging & Real-Time Notifications

Maintains an append-only transaction stream capturing all user activities, coupled with an in-app notification center notifying requesters of approval or rejection decisions.

5. Non-Functional Requirements (NFRs)

Category	Specification & Quality Standard
Zero-Backend Architecture	100% client-executable static web app; zero cloud database dependencies.
Data Persistence Engine	Persists all mutations (users, records, requests, logs) via LocalStorage serialization.
Responsive Layout	Fluid viewport adaptation across mobile (360px), tablet, and desktop (4K).
Privacy & Sanitization	Removal of proprietary logos, student/staff PII, and sensitive records.
Performance Benchmark	Sub-100ms view switching; sub-300ms debounced search filtering; zero latency.
Accessibility Standards	High-contrast text badges, keyboard-accessible modals, WCAG 2.1 AA compliance.

6. Success Metrics & Evaluator KPIs

< 15s
Evaluation Cold Start
Instant access via one-click credentials without setup.

100%
Workflow Pass Rate
Flawless upload, request, approve, and export cycle.

0 Loss
Storage Durability
Zero data degradation across page reloads & sessions.